The spectrum evidence supports partial redundancy of W1 under the retained strong augmentation policy. It does not establish a universal causal mechanism: the result is tied to the recorded seed-42 model, augmentation implementation, and paired diagnostic sample.

Feature-domain FDDEM remains a separate experimental direction; no final fully audited checkpoint-result pair was retained, so no numerical FDDEM outcome is added here.

Appendix G - Evaluation Metric Definitions

This appendix consolidates metric definitions used to interpret classification and segmentation results. Metrics from different tasks are not ranked against one another.

For N single-label classification samples, accuracy is

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\hat{y}_i = y_i]. \tag{G.1}$$

For class k,

$$\text{Precision}_k = \frac{TP_k}{TP_k + FP_k}, \quad \text{Recall}_k = \frac{TP_k}{TP_k + FN_k}, \quad F1_k = \frac{2\,\text{Precision}_k \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}. \tag{G.2}$$

Macro-F1 assigns equal weight to the K classes:

$$\text{Macro-F1} = \frac{1}{K}\sum_{k=1}^{K} F1_k. \tag{G.3}$$

Cohen's Kappa is

$$\kappa = \frac{p_o - p_e}{1 - p_e}. \tag{G.4}$$

For segmentation,

$$\text{IoU} = \frac{|M_{pred} \cap M_{gt}|}{|M_{pred} \cup M_{gt}|}. \tag{G.5}$$

Average precision integrates the precision-recall curve for one class. mAP50 averages class AP at IoU 0.50; mAP50-95 averages AP over IoU thresholds 0.50 through 0.95 in increments of 0.05. Full-test evaluation includes all designated images, whereas labeled-only evaluation restricts the set to images containing at least one BG or WSSV mask.

Healthy false-positive rate and disease missed-image rate are

$$\text{FPR}_{healthy} = \frac{\#\{\text{Healthy images with at least one retained disease prediction}\}}{\#\{\text{Healthy images}\}}, \tag{G.6}$$

$$\text{MissRate}_{disease} = \frac{\#\{\text{diseased images with no retained prediction}\}}{\#\{\text{diseased images}\}}. \tag{G.7}$$

Mask-count mean absolute error is

$$\text{MAE}_{count} = \frac{1}{N}\sum_{i=1}^{N} |\hat{n}_i - n_i|. \tag{G.8}$$

The project-defined Healthy-aware score is

$$\text{HScore} = \text{mAP50}_{labeled} - 0.05\,\text{MAE}_{count} - 0.15\,\text{MissRate}_{disease} - 0.10\,\text{FPR}_{healthy}. \tag{G.9}$$